

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

WINTER SEMESTER, 2024-2025
FULL MARKS: 120

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Explain the rules for identifiers in C. Provide at least 4 different examples of invalid identifiers and explain why they are invalid. 4 + 4
(CO1)
(PO1)

- b) What output is produced by the code given in Code Snippet 1? Use '_' (underscore) to represent any whitespaces that appears in the output. 6
(CO1)
(PO1)

```

1 #include <stdio.h>
2
3 int main(){
4     printf("%-6.3d, %5d\n", 35, 1024);
5     printf("%12.5e\n", 30.253);
6     printf("%8.4f\n", 83.162);
7     printf("%.4g\n", 123.45678);
8     printf("%5.3o\n", 32);
9
10    return 0;
11 }

```

Handwritten annotations for Code Snippet 1:

- Line 4: $1024 \rightarrow 1024$, $35 \rightarrow 35$
- Line 5: $30.253 \rightarrow 3.02530e+01$
- Line 6: $83.162 \rightarrow 83.1620$
- Line 7: $123.45678 \rightarrow 123.4568$
- Line 8: $32 \rightarrow 40$

Code Snippet 1: Sample program for Question 1.b

- c) Write a C program that keeps taking integers as input until 0 is encountered (do not consider 0 as one of the inputs), and finally prints the largest and smallest integers among them. 10
(CO3)
(PO1)

- d) "One of the strengths of the C language is its flexibility, but simultaneously it is also one of its weaknesses" — explain this statement. 6
(CO2)
(PO2)

2. a) Write a program that reads an integer number (as a single integer, not individual digits separately) from the user and finds the difference with its reverse number. A number's reverse can be obtained by writing its digits in the opposite order. For example: reverse of 452 would be 254. 12
(CO3)
(PO1)

A few sample input-output pairs are given in Table 1.

reverse = 0
 $reversed = reversed * 10 + digit$
 $= 0 * 10 + 2 = 2$
 $= 2 * 10 + 5 = 25$
 $= 25 * 10 + 4 = 254$

Table 1: Sample input-output pairs for Question 2.a

Sample Input	Sample Output
452	198
121	0
21	9

Handwritten calculations for reverse of 452:

$$10 \overline{) 452} \begin{array}{r} 45 \\ 40 \\ \hline 52 \\ 50 \\ \hline 2 \end{array}$$

$$10 \overline{) 459} \begin{array}{r} 45 \\ 40 \\ \hline 59 \\ 50 \\ \hline 9 \end{array}$$

digit = $2 \times 10 + 0 = 20$
digit = $20 + 5 \times 10 = 70$

- b) Find the output of the code given in Code Snippet 2 and justify the result by showing in details whether each of the if conditions will be evaluated as TRUE or FALSE. 10
(CO2)
(PO2)

```

1 #include <stdio.h>
2
3 int main(){
4     int a=1, b=20, c=5, d=3;
5
6     if(a>=1 && b>c){ True ✓
7         printf("This\n"); ✓
8     }
9     if(b-- > 19 || c==d){ ✓
10        printf("is\n");
11    }
12    if(d<a || ++c == 6 && d<0c/2){ 5/2 = 2.5 = 2
13        printf("not\n"); 3 < 2 ✓
14    }
15    if(!(c<d && a<=d) || b<c){
16        printf("easy\n"); ✓
17    }
18 }

```

Code Snippet 2: Sample program for Question 2.b

2. (c) Write a C program that takes a positive integer as input and prints the factorial of that integer. Assuming that you used a short int data type to store the factorial result, determine the largest input for which the program correctly prints the factorial.

6 + 2
(CO3)
(PO1)

3. a) Refer to the variable declarations as given in Code Snippet 3:

```

1 #include <stdio.h>
2
3 int main(){
4     char c = '\2';
5     short s = 2;
6     int i = -3;
7     long m = 5;
8     float f = 6.5f;
9     double d = 7.5;
10 }

```

Code Snippet 3: Sample program for Question 3.a

Determine the value and data type for each of the following expressions:

- i. $c * i$
- ii. $s + m$
- iii. f / c
- iv. d / s
- v. $f - d$
- vi. $(int) f$

- b) Write a C program that asks the user to enter an integer n , and then prints the square of all odd numbers, say $oddSquares$, where $1 \leq oddSquares \leq n$.

10
(CO3)
(PO1)

Table 2: Sample input and output squares for Question 3.b

Sample Input	Sample Output
Enter the upper limit: 150	1 9 25 49 81 121

- c) What are meant by *lvalue* and *rvalue* in C? Explain with valid examples.

8
(CO1)
(PO1)

4. a) Refer to the erroneous code given in Code Snippet 4 and the expected input-output given in Table 3:

10
(CO2)
(PO2)

```

1 #include <stdio.h>
2
3 int main(){
4     int n;
5     scanf("%d",&n);
6     for(int i=0;i<n;i++){
7         for(int j=0;j<n;j++){
8             if(j%2!=0)
9                 continue;
10            printf("%d",j);
11        }
12    }
13 }
```

Code Snippet 4: An erroneous program for Question 4.a

Table 3: Sample input and Expected output for Question 4.a

Sample Input	Expected Output
3	1 333
6	1 333 55555
8	1 333 55555 7777777

The program was supposed to take a positive integer n as input and the expected output of was to print all odd numbers from 1 to n in separate lines. Also, each number would be printed as many times as its own value. Find the errors in the code given in Code Snippet 4 and fix the code with minimal changes.

- b) "In C language, $v += e$ is not equivalent to $v = v + e$ " — justify this statement.

8
(CO1)
(PO1)

- c) Write a C program that asks the user to enter a date in the format dd-mm-yyyy, and then prints the date in the new format as shown in the sample input-output pairs in Table 4. Assume that the input is always a valid date. You must use switch-case to solve this problem.

12
(CO3)
(PO1)

Table 4: Sample input-output pairs for Question 4.c

Sample Input	Sample Output
11-12-2013	the 11th of December, 2013
31-05-2024	the 31st of May, 2024
03-02-2001	the 3rd of February, 2001

make sure to write -
return 0;
}